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CAUSES UNDERLYING THE HISTORICAL DECLINE IN EASTERN OYSTER (*CRASSOSTREA VIRGINICA* GMELIN, 1791) LANDINGS

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ABSTRACT Landings of eastern oysters (*Crassostrea virginica*) along the United States East Coast (mainly Rhode Island through South Carolina) increased to nearly 27 million bushels by 1890s, but they then declined by nearly 60% to 11.5 million bushels by 1940 and by almost 99% to 0.35 million bushels by 2004. Though overharvesting usually has been cited as the primary factor for the decline in landings from 1890 to 1940, the principal causes were: (1) a fall in demand for oysters, because consumers became aware that oysters could contain pathogens and competition from other foods increased in markets; (2) three economic depressions; and (3) biological and physical damage to the oysters and their beds (predation, siltation, severe storms, channel dredging, and harvesting by dredges). As a consequence of a huge decline in oyster landings in Delaware and Chesapeake Bays during and after the 1960s because of high oyster mortalities caused by the diseases MSX and Dermo, space became available in the United States national oyster market. The oyster industries of Prince Edward Island, Long Island Sound (Connecticut and New York), and the United States Gulf Coast took advantage of this and increased their production.

KEY WORDS: Oyster, *Crassostrea virginica*, East Coast, Gulf Coast, Canada, landings, consumer demand, pathogens, economic depressions, bed damage

INTRODUCTION

In recent decades, a topic widely discussed publicly and in estuarine science journals has been the decline in landings and low abundances of eastern oysters (*Crassostrea virginica* Gmelin, 1791) and the environmental and economic consequences in United States East Coast estuaries, especially Chesapeake Bay. The decline began during the 1890s and, except for some small upturns, has continued to the present. In 1890, total oyster landings were at their peak: 27 million bushels (120 million pounds of meats). The US population had grown to 63 million people in that year (Haines & Sutch 2006); oysters were a common food (Ingersoll 1881, Ingersoll 1887, McCutcheon 1993). In 2004, the oyster landings were only 0.35 million bushels, or 1.3% as large as in 1890 (Lyles 1969, NMFS landings statistics). Many scientists and managers have attributed the cause of the decline up through 1940s mainly to overharvesting, which supposedly depleted the beds of oysters and degraded the oysters' habitats to the extent that the beds had less and less capacity to produce oysters. Since the late 1950s, high oyster mortalities caused by the diseases MSX (*Haplosporidium nelsoni* [Haskin et al. 1966]) and Dermo (*Perkinsus marinus,* Levine 1978) have been included as another substantial cause of the decline (Galtsoff 1964, Kennedy & Breisch 1983, Hargis & Haven 1988, Rothschild et al. 1994, Paynter 1996, Smith et al. 2005).

Eastern oysters have several features that make them a good commercial product. They have a large biotic potential that provides them with the capacity to maintain large, viable populations in certain areas where habitat processes have not been disturbed. They grow on bottoms that are inexpensive for states and companies to use, they produce many larvae that abundantly populate bottoms that contain adequate substrates, the oysters can live for several or more years, sedentary oysters grow to market size in two to four years depending on latitude, and they acquire all their food from the estuarine waters. Moreover, eastern oysters are relatively easy to harvest in large quantity, and consumers consider them to be nutritious and tasty.

The present paper concentrates on the 50 y from 1890 to 1940, when oyster landings declined by nearly 60% to 11.5 million bushels. A series of factors will be described that, when taken as a whole, were far more important than overharvesting in causing the decline. The most prominent was a falling consumer demand for oysters: The demand for oysters was not consistently strong as has been assumed. The drastic decline in oyster landings in Delaware and Chesapeake Bays between the late 1950s and the present, that provided a stronger market and increased landings for the eastern Canadian, Connecticut and Gulf of Mexico oysters, is described.

A substantial amount of obscure published information and the general knowledge within the oyster industry and some state management agencies were overlooked by writers describing this decline. Overharvesting was rarely mentioned among the causes in these sources. A more accurate understanding of how various factors affected the oyster industry through time can be useful in generating plans to restore the habitats and oyster industry in the East Coast estuaries. See Luckenbach et al. (1999) for a review of oyster restoration efforts.

METHODS

Information on market demand for oysters during 1890 to 1940 was collected from articles in The Fishing Gazette. Printed weekly, 1884 and to the 1960s, this journal provides many details about the United States commercial fishing industry, including the oyster industry. Only a few United States libraries contain most of the issues. Siltation and its role in limiting the productivity of oyster beds was the subject of several articles in The Fishing Gazette and papers by Ingersoll (1881, 1887), Lee (1914a, 1914b) and MacKenzie (1975, 1983, 1996). Whitney (1994) and O'Connor et al. (1998) have discussed siltation further. Landings statistics are from various publications and

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annual landings statistics prepared by NOAA's National Marine Fisheries Service available on the internet.

The author, a shellfish biologist with the NOAA's National Marine Fisheries Service, worked closely with oyster fishermen and resource managers in Prince Edward Island (P.E.I.), Canada, and several states on the United States Atlantic Coast and on the Gulf of Mexico and also in eastern Mexico to examine factors related to oyster production over a 40-plus-years career (1960s–2007). Scuba observations were included in some investigations. Discussions regarding oystering issues were held with fishermen, company representatives, state biologists and officials, and university biologists (MacKenzie 1975, MacKenzie 1977, MacKenzie 1981, MacKenzie 1983, MacKenzie & Wakida-Kusunoki 1997).

PRODUCTION DECLINES IN INDIVIDUAL REGIONS

Long Island Sound

The Connecticut-New York-Rhode Island oyster industry landed about 3.8 million bushels of oysters in 1890, but landings declined after 1910. By 1920, the landings were about 2 million bushels. By 1940, landings were down to 1.5 million bushels. The private beds, which produced nearly all the oysters in Connecticut, were never overfished (pers. observ.).

Delaware Bay

Between 1890 and 1931, landings of market oysters in Delaware Bay were mostly between 1.5–2.4 million bushels. Between 1932 and 1940, production remained fairly constant at between 1.0–1.5 million bushels/yr (Ford 1997).

Chesapeake Bay

In 1890, about 20 million bushels of oysters were landed in Chesapeake Bay (Maryland and Virginia), but afterward its landings declined. By 1910, Chesapeake Bay oyster landings were 12.7 million bushels. In 1940, the landings were 7.3 million bushels.

CAUSES OF DECLINE IN OYSTER LANDINGS, 1890 TO 1940

In the eastern United States, a series of adverse conditions during 1890 to 1940 led to the decline of oyster production. The leading causes were: (1) a falling consumer demand for oysters, in part, because illnesses resulted from eating oysters contaminated with disease-causing bacteria, (2) increasing competition from other foods, (3) nation-wide economic depressions, (4) periodic losses of labor, (5) the Prohibition Era, (6) biological and physical degradation of oysters, beds and industry property and (7) neglect of seed beds.

FALLING DEMAND FOR OYSTERS CAUSED BY POLLUTION SCARES

During 1890 to 1940, the consumer demand for oysters was weakening and prices were low. The landed prices (corrected for inflation) for oysters remained nearly constant at roughly \$0.07/pound of meats or about \$0.40/bushel, and the landings of oysters declined by slightly more than half (Fig. 1). This resulted in poor economic conditions throughout the oyster industry, especially because production costs were rising (Killian 1918).

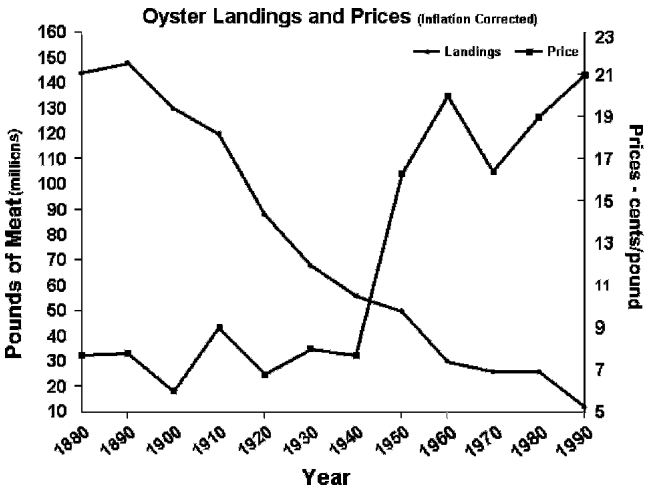


Figure 1. East Coast landings and prices (inflation-corrected) of oysters, 1880 to 1990. Data from Lyles (1969) and NMFS landings statistics.

Bacterial Pollution

In the 1800s, it was not widely known that oysters growing on beds close to towns and cities could contain typhoid (*Salmonella typhi* [Schroeder 1886] Warren & Scott 1930) and other disease-causing bacteria and viruses. Serious illnesses were common, but their causative sources were poorly known (Von Nyvenheim 1918, MacKenzie 1996). From the early 1900s through the 1920s, flush toilets were installed in most homes and apartments (MacKenzie 1992). Many toilets drained into sewerage lines that emptied into the local rivers and bays where oysters were harvested (Fig. 2).

In the early 1900s, the United States was developing a great concern for improved sanitation regarding foods, or, as some termed it, "a pure-food hysteria." The foods included oysters, probably because a great oyster scare in England in 1902 became known in the United States. In Emsworth Harbor, England, sewage-polluted oysters infected about 20 people with typhoid at a banquet and three died (Anonymous 1903). Many people in the United States were becoming ill with typhoid and gastrointestinal disorders caused by the consumption of raw and undercooked oysters. Health officials and journalists vented a flood of criticism about oysters being harvested from polluted waters and also handling practices within the oyster



Figure 2. A large sewer emptying untreated sewage into a river. This sewage flows over extensive shellfish grounds (Turner 1911).

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industry. Newspaper accounts with titles such as "Pollution of Oysters" and "Typhoid Fever and Oysters" promoted the idea that oysters were a risky food to eat (Anonymous 1911), and some newspapers and journalists (Turner 1911) targeted the oyster as the main focus of their attacks on the American food system (Figs. 2, 3). Oyster sales plummeted (Kochiss 1974, MacKenzie 1996).

In 1906, the United States Congress passed several pure food laws, which regulated the entire food industry. Oyster beds were examined regularly and they were closed to harvesting if the oysters and water contained hazardous bacterial contamination. The laws brought major changes in handling, packing and shipping of other foods, too, including milk, ice cream and candy (Anonymous 1909). Processing methods used by the oyster industry since it began were found inadequate. By 1909, the health boards condemned the industry practices of shipping fresh oyster meats in wooden buckets and barrels, shipping the meats in direct contact with ice, and adding water to the meats (Anonymous 1909). The ice was harvested during winter from rivers, lakes and ponds, some of which may have been contaminated (Weightman 2003).

In 1907, H. C. Rowe organized the Oyster Grower and Dealers Association whose aim was to save the industry, then suffering from economic decline (Anonymous 1907). The association took steps to restore public confidence in oyster purity through advertisements, but efforts were not intense enough to be of much help. In 1911, Rowe said the industry had continued to decline. Weak demand, low prices, and increasing costs of fishing and packing resulted in smaller profits for owners, marginal incomes for company workers on boats and in plants, and for the fishermen harvesting oysters on public beds (Anonymous 1917). Oysters often were delivered to customers in poor quality caused by delays in transportation and in some cases they were poorly shaped and undersized (4–5 cm long) often with barnacles and small mussels attached. Whenever customers purchased them and found them to be less than satisfactory, they tended to seek different foods thereafter (Anonymous 1905d, Anonymous 1914, Anonymous 1917, Nichol 1937).

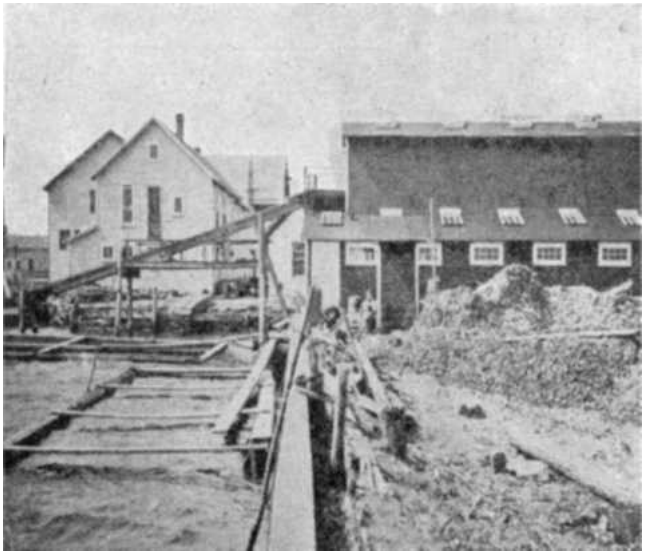


Figure 3. Oyster floats, left foreground, located near a shucking house. The buildings in the foreground have a privy used by 100–150 men. The drainage from it seeps through the loose soil and a porous wooden wall into the water, which bathes the floats. Drainage from decaying shell heaps and the washings from the oyster house likewise flow into the same water. Analysis of this water and oysters on the floats showed contamination (Turner 1911).

Closures of Marketing Areas

The spread of bacterial pollution forced state health officials to rule that some beds, mostly those near human population centers, could no longer be used for harvesting oysters destined for direct sale to consumer markets. The practice of "giving oysters a drink" was also banned. The bed closures occurred over the entire range of the eastern oyster from the Canadian Maritime Provinces to the states in Eastern Mexico, including states from Massachusetts to Texas in the United States. The convenient, quick and inexpensive transport of oysters to nearby markets had to be abandoned. An example from Connecticut will illustrate some difficulties this presented to the oyster industry.

Health officials ruled that Connecticut's New Haven beds could not be used any more for harvesting oysters to be sold directly to consumer markets. They could be used only for producing seed oysters. The oyster companies then had to transplant their seed oysters to acceptable grounds for further growth and eventual marketing. The grounds were located farther west in Connecticut; and in Long Island, NY; Narragansett Bay, RI; and Wellfleet, MA. The companies also reduced the sizes of their large New Haven shucking operations that had been described about 30 y earlier by Ingersoll (1881). The companies then had far fewer shells available to spread on their beds to collect more crops of seed oysters (MacKenzie, 1996). Furthermore, sales of oysters slowed as newspapers were reporting that oysters were not consistently wholesome, and also because cheaper oysters from other states were shipped into Connecticut's markets. Adding to the burdens of the companies, state taxes on the Connecticut leased beds were raised. The overall effect was to weaken the Connecticut oyster industry so much that some companies abandoned their operations during the early 1900s (Rowe 1912, 1914, Anonymous 1912).

In Rhode Island, Raritan Bay and Delaware Bay, many oysters were carried in from the beds and held in brackish waters on floats or on the intertidal bottoms of creeks for a day or longer before being sent to markets (Anonymous 1905a, Ford 1997) (Figs. 3 and 4). The purpose was to swell the oysters' meats in the fresher water and also allow the oysters to remove mud and sand from their shell cavities (Nelson 1911). Because this "giving oysters a drink" process was commonly carried out near human population centers, some outbreaks of typhoid among consumers were traced to these oysters (Anonymous 1905a, Churchill 1921). Authorities eventually banned the practice and the industry switched from this to holding shucked meats for 15–20 min in tanks of circulating freshwater (blowers) to accomplish the same ends (Nichol 1937).

Typhoid Outbreak in 1924

The oyster industry received a huge setback in 1924, when most national newspapers and radio broadcasts (from the small number of radios present at this time) reported that many people had become ill and some died from typhoid after eating

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Figure 4. Removing oysters from a creek in Keyport, New Jersey, about 1910. The oysters were "given a drink" in the brackish water for 1–2 days (Source, Steamboat Wharf Museum, Keyport, NJ).

oysters (Kochiss 1974). Those affected in Chicago had eaten oysters harvested in Raritan Bay, New York and New Jersey. Oyster sales nationwide plummeted by 50–80%, as consumer suspicion was cast over all oysters (McCarthy 1925). The negative effect on the oyster industry was felt at least from Long Island Sound to South Carolina (Chestnut 1951, Maggioni & Burrell 1982, MacKenzie 1996, MacKenzie 1997, Ford 1997, Burrell 2003).

FALLING DEMAND FOR OYSTERS CAUSED BY COMPETITION FROM OTHER FOODS

Wholesome foods in increasing variety, volume and convenience at affordable prices became available to consumers beginning in the late 1800s and increasingly so during the 1900s. Fresh beef, pork, and canned vegetables and fruits were available in markets year round. For example, in 1869, the Campbell Soup Company began to produce canned tomatoes, vegetables, and soups (including oyster stew). Some smaller companies that produced the same products formed a little later. In 1897, the Campbell Soup Company invented condensed soup, using a process that eliminated most of the water. The costs of packaging, shipping and storing it were much lower (Shea & Mathis 2002). Such foods were widely advertised and they became staples in consumers' diets.

Beginning around the 1920s, shrimp landings in the South Atlantic and the Gulf of Mexico sharply increased (NMFS landings statistics); and since the early 1930s, when otter trawls became common, large quantities of fresh finfish from the offshore fishing grounds became available for consumers (Shea & Mathis 2002). Shrimp and finfish, less available to compete with oysters in markets before this, became popular with consumers (Nichol 1937).

Table 1 lists the annual per capita consumption of meats, finfish and oysters from 1880 to 1940. Between 1910 and 1940, the annual per capita consumption rate remained steady at an average of about 40 pounds each of beef and pork, 4 pounds of lamb, and 10 pounds each of chicken and fish; whereas the annual consumption rate of oysters declined, from 3 (in 1880) to 0.6 pounds per capita. Safety and greater ease of handling beef, pork, chicken and finfish and the opportunity for increased variety gave them selling points over oysters. The oyster consumption figures include the entire United States. Far more oysters were eaten in communities on the East and Gulf Coasts. For example, in 1907, oyster consumption in New York City averaged about 0.5 pounds/person/week (Anonymous 1907).

TABLE 1.

The annual/capita consumption of meats, fish, and oysters (in pounds) in the United States, 1880 to 1940 (Anonymous 1970, Lyles 1969).

Year	Beef	Pork	Chicken	Lamb	Fish	Oysters
1880	—	—	—	—	—	3.0
1890	—	—	—	—	—	2.7
1900	—	—	—	—	—	2.0
1910	48.5	38.2	11.0	4.2	8.7	1.6
1915	38.8	40.9	10.7	4.0	10.9	1.3
1920	40.7	39.0	9.7	3.6	9.7	1.0
1925	41.0	41.0	10.1	3.4	9.9	0.6
1930	33.7	41.1	11.1	4.4	9.2	0.7
1935	36.6	29.7	9.3	4.8	10.1	0.6
1940	37.8	45.1	10.0	4.3	10.9	0.6

ECONOMIC DEPRESSIONS

Each of three major economic depressions in this period adversely affected oyster sales. During a major nation-wide economic depression in the early 1890s, NY City dealers had a glut of oysters on hand. Working-class people who previously had consumed most of the oysters could no longer afford them (Anonymous 1893).

The second such depression came in 1907 to 1908. Lee (1914a) reported that oysters on vast areas of private grounds in Virginia had been held over year by year with limited sales.

The third depression began with the 1929 stock market crash and continued through the 1930s. Considering only Connecticut, New York, New Jersey and Chesapeake Bay (Maryland and Virginia combined), from 1929 (little effect as yet) to 1932 (the lowest period), oyster landings fell in each location by 3.1%, 34%, 57% and 15.8%, respectively. For the same areas and the same time period, landed prices fell by 10%, 37.4%, 42.5% and 47% (Lyles 1969).

PROHIBITION ERA

Some oysters were consumed in saloons before the 1920s. Patrons were offered free or inexpensive oysters on the half-shell and in stews to bring them into their establishments where they were sold beer and other alcoholic drinks. When the saloons were closed during the Prohibition Era, 1920 to early 1930s, this oyster market was mostly lost.

PERIODIC SHORTAGES OF LABOR

At times, the number of laborers available to work on oyster boats and in packing plants was too small for efficient oystering operations. This occurred when many men went into the armed services and to work in war industries during World War I (1917–1918), and when economic conditions were good and jobs more desirable than oystering were available (the 1920s, for example) (Anonymous 1905a, Chestnut 1951, Burrell 2003).

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BIOLOGICAL AND PHYSICAL DEGRADATION OF OYSTERS AND BEDS

The condition of many beds became marginal for producing oysters. The beds commonly contained predators, which destroyed many seed oysters, and they were subjected to such negative physical factors as siltation, severe storms, reductions in river flows, dredging of navigation channels and damage from oyster dredging. Each factor reduced the quantities of oysters the beds could support.

Consistent Predation on Oysters

Oyster drills, [*Urosalpinx cinerea* (Say, 1822) and *Eupleura caudata* (Say, 1822)], preyed extensively on seed oysters on private beds and they prevented public oyster beds from expanding to locations where salinities were >15–18 0/00 in about all East Coast States. The drills caused heavy oyster mortalities in Connecticut (MacKenzie 1981), Delaware Bay (New Jersey) (Ford 1997) and the higher-salinity, oyster-growing grounds in Virginia (Haven et al. 1978). The starfish (*Asterias forbesi* [Desor]), whose distribution is also limited to salinities >15 0/00, was a major oyster predator in Long Island Sound (Galtsoff & Loosanoff 1939). Other oyster predators included crabs and drumfish.

Siltation

The accumulation of silt deposits on oysters and shells prevents oyster larvae from setting. Siltation has been mostly overlooked as a factor that degrades the oysters' environment, because it could not be observed by fishermen and scientific observers in boats. The fishermen included tongers, who could see silt on oysters when they lifted them to the surface, and boat captains, who had to wash mud from the oysters in their dredges, but they were not aware of bottom conditions as silt overlaid setting surfaces on the bottom (Fig. 5). Silt deposits on shells can be observed on intertidal oyster beds (Maggioni & Burrell 1982).

When the Europeans first settled the East Coast, most of the land westward to the Mississippi River was covered with dense forests and shrubs. The land was steadily cleared to make way for farms and towns and to obtain wood for fuel and construction (Fig. 6). By the 1850s, the deforestation was especially prevalent in the East Coast States from southern Maine through South Carolina, and later the land farther west was mostly cleared. By 1920, the drainage basins of the Hudson River, Delaware River, Chesapeake Bay and the bays of North Carolina had been nearly denuded of forests; large areas were being used to grow crops and farm animals (Whitney 1994). The land surface was eroded by water runoff, especially in the spring. Annual soil loss on the forested lands was measured at <0.01 tons/hectare/yr; on the bare fallow land at



Figure 5. Deposits of silt on oyster shells, Parson Island, MD (from MacKenzie 1983).

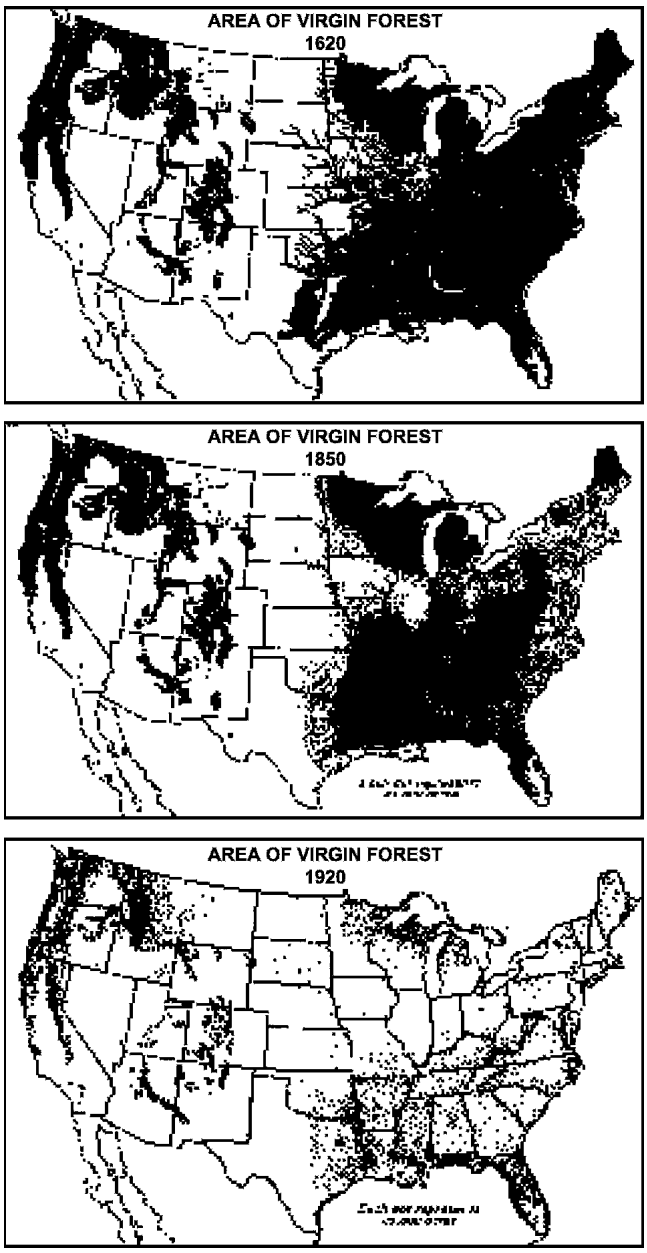


Figure 6. Extent of uncut virgin forest in the United States, 1620, 1850, and 1920. Areas of forests are based on estimates by states. Each dot represents 25,000 acres. Dots are not all correctly located from Whitney, 1994; originally from Greely (1925), Economic Geography 1:4–5.

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155 tons/hectare/yr; and on the land used for corn at 164 tons/hectare/yr (Bennett 1939).

Many streams and rivers carrying silt from the farm lands flowed into estuaries where the silt settled on oyster beds (MacKenzie 1983, MacKenzie 1996, Abbe 1988, Smith et al. 2005). The silting of oyster seed grounds was most prevalent on oyster beds in Delaware, Delaware Bay, and in Chesapeake Bay and its tributary rivers, including the James River, but it also occurred in Connecticut and P.E.I. and other locations (MacKenzie 1983). Silt also settled in channels and severely shoaled their depths, resulting in the need to dredge them frequently (Brush 1986, Whitney 1994).

Some of the earliest writers who described the oyster industry had observed how siltation affected oyster beds. Ingersoll (1881, 1887) believed that siltation reduced the productivity of oyster beds in Maine after forests were removed and the bare land was exposed to rains. Lee (1914b) reported that spring freshets covered Virginia's James River seed beds with mud and silt. He suggested that the tongers should be allowed to harvest seed from the beds as late as June because tonging washes the silt off the bottom and currents disperse it, allowing oyster larvae to set. This recommendation was followed and the James River consistently produced at least 2 million bushels of seed oysters a year thereafter until 1960–1961 (D. Haven, pers. comm.).

The heaviest silt loading in the various drainage basins occurred during the 1880s and 1890s. During the 1900s, the siltation may have gradually declined as the drainage basins became more forested because of reduced farming and logging (Whitney 1994, O'Connor et al. 1998).

Adverse Weather

Severe wind storms and hurricanes have buried oysters and also washed those on shallow beds onto nearby beaches, leaving the beds barren. A few examples can be cited. In 1914, the fishermen harvesting oyster seed on the public beds (water depth, 1.5–2.4 m) in New Haven Harbor, CT, had to abandon the beds because a severe storm had washed most of the oyster seed onto a nearby beach (Anonymous 1914). In 1910, a hurricane damaged the South Carolina oyster industry (Burrell 1982, Burrell 2003); the Hurricane of 1933 damaged oyster beds in North Carolina (Chestnut 1951); and the 1938 Hurricane buried nearly all the oysters and shells and destroyed many boats, wharves and packing plants in Long Island, New York, Connecticut and Rhode Island (MacKenzie 1996).

The oyster industry found that restoration after a hurricane was costly. It required rebuilding shore facilities, salvaging boats, bringing the beds back into condition to produce oysters, and attempting to return oysters to traditional markets before their customers switched to alternate foods (H. Bloom; J. R. Nelson, pers. comm.). In Connecticut, at least, the restorations were only partial; its industry was smaller after the 1938 Hurricane (MacKenzie 1996).

Extremely cold winters also hampered the oyster industry operations. The winter of 1904–05 was so cold that Long Island Sound had large amounts of ice. Nearly all the hundreds of stakes the Connecticut oystermen had in place to mark their grounds were removed by the ice as it drifted. The stakes had to be replaced, using unprepared for effort and cost, by the spring of 1905 (Anonymous 1905b). The same winter also resulted in a poor oyster season in Virginia. Its fishermen were prevented from working and planters had few oysters to sell for almost 2 mo (Anonymous 1905c).

Reductions in River Flows

Most oyster beds occupy the sections of rivers and bays, where water salinities range from about 5–18 0/00 (Galtsoff 1964). As towns and cities grew, they removed water in increasing quantities from the rivers that flowed by them, and also some river flows were diverted (SC Water Resource Comm. 1979, Bradley et al. 1990). The consequence was less freshwater flowing into the estuaries. The positions of the oyster beds then became relocated upstream where the areas of suitable oyster habitat usually were smaller, and, being closer to human population centers, they could be polluted. The oysters became scarce on the lower extremities of beds, because fouling organisms usually covered the shell substrates preventing oyster larvae from setting and predators destroyed the seed oysters in waters >15 0/00 (Dugas et al. 1997).

Channel Dredging

Some oyster beds in most every estuary were partially destroyed when channels were dredged through them from deep waters to ports for navigational purposes, and construction of the Intracoastal Waterway also destroyed some shellfish beds (Burrell 2003). The dredging process also releases silt into the water (Maggioni and Burrell 1982). Channel dredging also may have allowed high-salinity wedges to penetrate into oyster beds, bringing predators with them (Dugas et al. 1997).

Habitat Destruction by Oyster Dredges

Oyster harvesting with dredges sometimes removed so many oysters and shells that beds were left with insufficient substrate for ready-to-set oyster larvae. Ingersoll (1881) had reported that some Chesapeake Bay oyster beds were partially depleted of oysters after many years of harvesting with dredges from about 1850 to the 1870s. Stevenson (1894) reported that by 1890 at least 1,000 sailing vessels were dredging oysters in Maryland.

Winslow (1884) wrote that the oysters in Chesapeake Bay were declining in abundance because of the use of dredges and overharvesting. He based his statement on the declining ratio of bushels of oysters to bushels of shells collected in dredged samples over a period of years: 3.7:1 in 1876, 1.9:1 in 1879, and 1.3:1 in 1882. Winslow stated that the only new factor in the oysters' habitat was dredging and overharvesting, whereas everything else had remained stable. But is this true? Harvesting likely was a factor here, but Winslow was viewing oysters that had been dredged or tonged onto the deck of a boat. Neither he nor Ingersoll were aware of how siltation might have reduced the sets of oyster spat on the shells and oysters and contributed to the decline. Regardless of the siltation, the oyster harvesting rate would need to have been extremely low for the 3.7:1 ratio of oysters to shells to be sustained year after year.

The beds from which market-sized oysters were harvested by dredges could not produce a consistent yield of oysters when they were accumulating silt. Fishermen dredged for market oysters mostly from September through December, whereas most silting occurred during March and April snow melts and rains. The result was fewer clean shell surfaces available as

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cultch for oyster larvae on the dredged bottoms. Little evidence exists that the declining oyster stocks in estuaries reduced the numbers of oyster larvae produced (Fig. 7).

Moore (1911) observed the depletion of cultch on the public seed beds in Delaware Bay (Delaware) during the spring harvesting of seed oysters. A cull law was not in effect at the time. The crews retained the shells to spread with the seed on their oyster growing beds farther down the bay. The shells served to harden their bottoms. Moore (1911) examined the condition of the seed beds after the dredging. In patches, the shells and seed were completely removed, leaving a surface of muddy sand. Oyster larvae are unable to set on such a surface, and in succeeding years fewer oysters would be present on the beds were the practice to continue.

Dredging can also level the surface of an oyster bed. On beds consisting of packed deposits of oyster shells, dredging tends to leave the bed surface fairly level with little protruding shell. But bottom surfaces harvested with tongs are rougher with protruding shells, in part because the action of the tongs gouges some shells out of the bottom. Such a rough surface has more available setting sites and more oysters than a flat level surface (personal observations in Mississippi, 1975).

In some areas, seed oysters once grew on the surfaces of low mounds of oyster shells. Over time, as dredge boats harvested the seed and some shells with them, the mounds were removed and the sand or muddy-sand substrate beneath them was left with a partial shell cover and less attachment area for oyster larvae (personal observations in Delaware Bay, New Jersey).

NEGLECTING SEED BEDS

In the 1800s and into much of the 1900s, most shells from oyster shucking plants, New Jersey through Virginia, were not replanted onto beds to collect sets of seed oysters. The shucking plants owned the shells and they sold most for uses on land. The shells were used to produce chicken scratch and lime for the agricultural industry, and for surfacing roads in towns and in forests (for hauling logs), and as railroad beds, land fill and, to a small extent, ballast in ships. In the 1870s, farmers were using the oyster shells as chicken scratch on a small scale (Ingersoll 1881), but, around 1890, they were using increasing quantities (Stevenson 1894). State governments purchased relatively small amounts of shells to spread on their public seed grounds (Galtsoff 1943, Ford 1997), because they apparently did not believe it worthwhile to spend much money on an industry that was generating little revenue.

During the 1930s, Fiedler (1936) listed the quantities of shells that Maryland and Virginia oyster-shucking plants sold as chicken scratch and agricultural lime. In 1935 (a typical year), Maryland oyster landings were about 2.5 million bushels (Lyles 1969). The oysters sold for $1 million, or $0.42/bushel. In that year, about 4 million bushels of shells were sold to make chicken scratch and lime. The shells sold for $213,777, or $0.053/bushel. About 325,000 bushels of shells were planted on Maryland oyster beds, or about 7.5% of its shell supply. In 1935, Virginia oyster landings were 2.3 million bushels valued at about $1 million, or $0.44/bushel. About 2.5 million bushels of shells were sold to make chicken scratch and agricultural lime, for $220,000, or $0.09/bushel. Virginia planted 260,000 bushels of shells on its beds, or 9.5% of its supply (Fiedler 1936). Shells sold for building roads or other uses in Maryland and Virginia were not included.

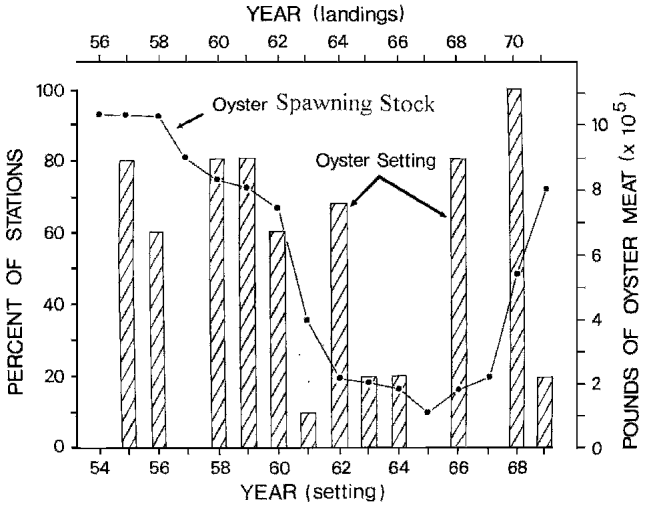


Figure 7. A comparison of the setting potential of oyster spat (histobars) with the size of the spawning stock (solid line) in Connecticut, 1954 to 1969. Landings data are lagged two years on setting data. The percentages of 10–12 stations of test shells having successful commercial oyster sets (i.e., at least 10 spat per shell/season) show the setting potential of spat (from MacKenzie 1983).

FACTORS THAT BENEFITED THE OYSTER INDUSTRY

Some factors benefited the oyster industry during 1890–1940. They were developed by the states, the industry and the federal government, whereas some, such as manufactured ice, gasoline and diesel engines, metal containers and trucks, were from other sources. They helped to modernize oystering as a producer of food for the public and may have helped to slow the decline in landings.

State, Industry and Federal Actions

In the late 1800s and early 1900s, several states passed cull laws: Oystermen who harvested oysters from public grounds had to return the empty shells to the bottom for the purpose of maintaining adequate substrates for setting oyster larvae (Killian 1918, Ford 1997). Without this regulation, most of the productive beds would have been depleted (Prytherch 1932). But saving the public grounds from ruin came at a cost to the oyster companies, because they had to hire extra men (at least 10 total men rather than about 6 men previously on each New Jersey vessel [Ford 1997]) to cull the seed. Also, their planting grounds then had to be hardened by shells that they previously had been able to sell, or else they were left in a marginal condition for growing oysters.

An industry action that helped was the formation of the Oyster Growers and Dealers Association. Comprised of growers from the East and Gulf coasts, the association gave the industry better capacity to deal with various types of issues, such as market promotion and government regulations (Anonymous 1911).

As a consequence of the 1924 typhoid outbreak, which was traced to sewage-polluted oysters, local and state public health officials and the shellfish industry became sufficiently alarmed

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over the outbreak to request that the Surgeon General of the United States Public Health Service develop necessary control measures to ensure a safe supply of oysters to the consuming public. In accordance with this request, the Surgeon General called a conference of representatives from federal and state agencies and the shellfishing industry in 1925 to deal with the problem. The members of the conference recommended some resolutions for sanitary control within the oyster industry. They included:

"The beds on which shellfish are grown must be determined, inspected and controlled by some official state agency and the U.S. Public Health Service."

"The plants in which shellfish are shucked or otherwise prepared or packed by the shipper must be inspected and controlled by some official state agency and the United States Public Health Service."

"There must be such governmental supervision and such trade organization as will make plain the source of the shellfish and will prevent shellfish from one source being substituted from another source. This will be chiefly a problem of the individual state."

"The methods of shipping must be supervised, inspected, controlled and approved by the proper official federal and state agency."

"The product must conform to an established bacterial standard and must meet federal, state, and local laws and regulations relative to salinity, water content, food proportion and conform to the Pure Food Laws standards."

To implement this program, the members of the 1925 conference agreed that the producing states would issue "Certificates," i.e., permits to operate, to shellfish shippers that meet agreed upon sanitary standards. The Public Health Service would serve as a clearinghouse for information on the effectiveness of the state control programs (Anonymous 2005).

Oysters became a safer food for people thereafter because they were likely to be free of harmful numbers of pathogenic bacteria and viruses. Consumers have remained skeptical, however, that oysters, especially eaten raw, are consistently healthful to eat.

In the mid 1930s, the industry in Oyster Bay, NY, developed a means of killing starfish by spreading of granulated quicklime over oyster beds for use in New York and Connecticut (Galtsoff & Loosanoff 1939). Quicklime effectively killed them when applications were 1,500 to 2,000 pounds/acre, but lighter doses were ineffective.

Useful Imports

Soon after 1900, a series of production factors aided oystering operations including marketing. Gasoline motors and propellers were installed in sailing boats to increase their usefulness (harvesting days had been lost during calm periods when dredging boats were using sails for propulsion), and dredges were hoisted by motorized winches rather than by hand winders (Anonymous 1912); the manufacture of clean ice was developed to replace the uncertain ice supply from freshwater sources; and metal shipping containers replaced wooden containers for oyster meats (Von Nyvenheim 1918). In the 1920s and 1930s, trucks became available to transport oysters, replacing horse-drawn wagons and eventually railroads. The use of trucks coincided with the construction of good roads.

EFFECTS OF THE DISEASES MSX AND DERMO ON OYSTER LANDINGS: LATE 1950s TO THE PRESENT

Most of the oysters approaching market size in Chesapeake and Delaware Bays have been killed by the diseases MSX and Dermo beginning in the late 1950s and continuing to the present. The oysters initially were killed by MSX (Andrews et al. 1962, Haskin et al. 1966, Andrews & Wood 1967, Haskin & Ford 1979, Haskin & Ford 1982, Ford & Tripp 1996). In the early to mid 1950s, the combined oyster landings in Delaware and Chesapeake Bays were 8–9.5 million bushels/year (Delaware Bay, 25%; Chesapeake Bay, 75%). From 1962 to 1980, they fell by 50%, but stabilized at 4–5 million bushels/year. They then fell sharply and in 2004 about 75,000 bushels were landed (NMFS landing statistics).

RISES IN OYSTER LANDINGS IN THREE REGIONS BEGINNING IN THE 1970s

The steep decline in oyster landings in Chesapeake and Delaware Bays provided marketing opportunities for three other oyster-producing areas, because a large gap was left in the North American oyster market in the 1960s and 1970s and thereafter. P.E.I. and Connecticut found that the landed prices of their oysters were rising sharply, and they could sell far more oysters. In addition, the oyster industry in the Gulf of Mexico States was able to double its production by selling far more oysters to markets along the Atlantic Coast. The MSX and Dermo diseases did not kill oysters in P.E. I. and Connecticut. Dermo was present in the Gulf States but with good management the growers could limit the extent of oyster mortalities from this cause.

Prince Edward Island

Prince Edward Island (P.E.I.) oysters are eaten only on the half-shell. Their meats are thin compared with those of eastern oysters harvested in the United States: They yield too little meat when shucked for sales of their meats for cooked dishes to be profitable. Their size is good for eating them on the half-shell, they have an excellent salty flavor and they bring relatively high prices in the half-shell market. The fishery grades the oysters into four price categories, all based upon shell shape, which ranges from long and narrow, the cheapest, to more oval, the most expensive. Most oysters are harvested with hand tongs (MacKenzie 1975).

Before the 1970s, production of P.E.I. oysters was limited by supply and market availability. From 1955 to 1970, P.E.I. oyster landings were about 40,000 bushels/year, but in 1972 landings were 17,500 bushels (Jenkins et al. 1997). The demand for P.E.I. oysters then began to rise, landings rose sharply and in 2003 they had risen to about 180,000 bushels with a dockside value of about $10 million (Fig. 8) (R. Gallant, pers. comm.). The prices (in Canadian dollars) per bushel of landed oysters were about $1–2 in the 1950s, $7 in the 1960s, and $17–18 in the 1970s. They continued to rise and were $40 in 1985, $50 in 1990, and about $60 in the late 1990s and 2000s. About 30% of P.E.I. oysters have been shipped for sale to the United States in recent years (R. Gallant, pers. comm.).

In 1972, the P.E.I. Provincial government directed that efforts be made to increase the oyster quantities on the beds and consequent landings, when the oyster markets were

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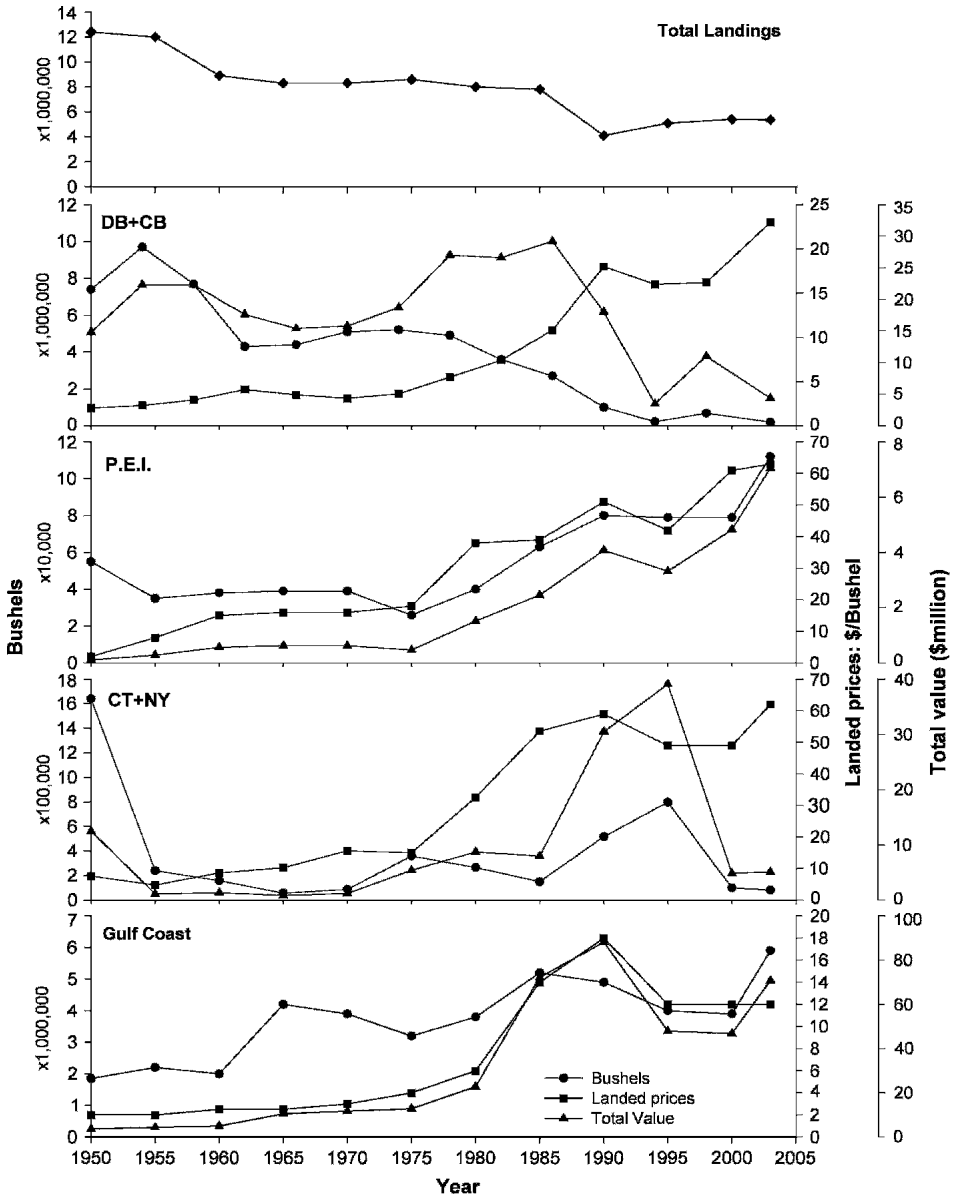


Figure 8. Oyster landings and prices from 1950 to 2003. The top panel shows total oyster landings in Prince Edward Island, Connecticut and New York, Delaware and Chesapeake bays, and the Gulf of Mexico. The four lower panels show landings, prices, and total landed values for individual areas in Delaware (DB) and Chesapeake Bays (CB); Prince Edward Island (P.E.I.); Connecticut (CT) and New York (NY); and the Gulf of Mexico.

expanding and prices were rising. The efforts implemented included transplanting oysters from grounds inaccessible to fishermen to good harvesting beds, mining and spreading shells on beds to collect seed oysters and controlling starfish with quicklime (MacKenzie 1975).

Long Island Sound

The increase in the Connecticut and New York oyster landings was caused by expanded culturing of beds to take advantage of the increased prices, which were roughly the same as those in P.E.I., and the larger markets (Fig. 8). After many years of lackluster efforts to produce oysters because market prices were low and profits marginal, the Connecticut companies began earnestly to produce more oysters beginning in the late 1960s. Their bed culturing involved spreading much larger quantities of shells to collect oyster seed, reducing mortalities of seed oysters caused by starfish and oyster drill predation and smothering in silt, and transplanting the seed oysters to growing and marketing grounds (MacKenzie 1981, MacKenzie 1996). Oyster landings reached rose to nearly 1 million bushels/yr in the early 1990s. But in the mid 1990s, the diseases MSX and Dermo began killing in Connecticut oysters and have since killed most market-sized oysters every year.

Gulf of Mexico

Production of Gulf Coast oysters has been limited by consumer demand throughout history. For example, LA oyster supplies have consistently far exceeded the capacity of US

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Figure 9. Scenes of the oyster industry in the United States Gulf of Mexico during 2002 to 2003. Upper left shows full oyster dredge being emptied, Pass Christian, MS; bottom left shows tonging boat with a large oyster harvest, Pass Christian, MS; upper right shows boatload of oysters, Galveston Bay, TX; and lower right shows tractor trailer truck with full load of sacked oysters, Empire, LA, being readied for transport to an East Coast market.

markets (interviews of several oyster wholesalers, fishermen, state oyster managers and scientists). Landed prices of oysters from the Gulf Coast and Delaware Bay-Chesapeake Bay were about the same from 1950 to 1990, with a wide range from $2–18/bushel. In contrast, from 1995 to 2003, landed prices of Gulf Coast oysters ($12/bushel) were below those of Delaware and Chesapeake Bay oysters.

In the last three decades, about half of the increased production of Gulf Coast oysters has been sold to East Coast States to meet the needs of their markets (Fig. 9). Some of the oysters have been shucked in East Coast packing plants and the meats sold in the Midwest States.

CONCLUSION

This history of causes underlying the downward trend in eastern oyster landings has shown that the strength of consumer demand played a huge role in determining the size of the landings. The landings increases in P. E. I., Connecticut, and the United States Gulf of Mexico also have shown that oyster landings can be increased in some circumstances when markets become larger when coupled with higher landed prices for oysters. Overharvesting oysters was not the main reason for the decline in landings. Moreover, reducing the rate of harvesting, *per se*, will not bring about a restoration of oyster abundances.

Oyster abundances in Connecticut, New Jersey, Delaware, Maryland and Virginia currently are relatively low, and hundreds of acres of their bottoms that once grew oysters have lain otherwise barren for many years. Future programs designed to increase the oyster abundances for the overall purpose of restoring benthic environments, controlling excess phytoplankton blooms (Newell 1988, Coen et al. 1999, Coen et al. 2007) and also reviving the states' indigenous oyster industries need to include enhancing the consumer demand for oysters to be successful.

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